

TAC459 - Team G Roadmap v3

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1. Product motivation and context

- Syllabus to Task List Converter: Students upload or paste their syllabus and the app extracts all deadlines, assignments, and exams into a clean structured list. Built with Claude AI (Anthropic) for high-accuracy extraction, with NLP concepts from TAC 459 (named entity recognition, date extraction) demonstrated in the companion Jupyter notebook.
- The Problem: Students receive 4-5 syllabi every semester with inconsistent formatting and buried deadlines. Manually copying dates into calendars is tedious and error-prone. Missing one deadline can seriously hurt a grade.
- Why It Matters: Every college student faces this every semester, massive and recurring user base. Students are already motivated to solve it, they just lack the right tool. Existing apps like Notion and Google Calendar still require manual data entry.
- The Opportunity: Similar applications (Due Gooder, Taskade, Syllabuddy, Sylcampus) are highly fragmented, hidden behind expensive paywalls, and aren't always effective. SyllabusSync v3 is a fully functional MVP with AI extraction, calendar export, and a polished UI — all in a single file with no installation required.

2. Target users and problems

- SyllabusSync is built for college students, but the problem hits different groups in different ways. There are 18-19 million college students in the US. Three segments:
 - Undergraduate students are the core user. They're juggling 4–6 courses at once with a high density of deadlines, and they're often most overwhelmed right at the start of semester — exactly when all the syllabi drop. They need something fast and zero-effort.
 - Graduate students balance coursework with research, TA duties, and advisor meetings. Their syllabi also tend to change mid-semester with no clear notice, making it hard to keep their task list current.
 - Transfer and international students are often unfamiliar with how U.S. syllabi are structured, and language barriers make manually scanning a 10-page PDF for dates even more painful. They stand to gain the most from automation.
- User flow: A student opens the app, uploads a PDF or pastes text, clicks "Extract with AI," reviews the extracted tasks (editing or deleting as needed), then exports directly to Google Calendar via .ics file or downloads a CSV.
- 5 Pain Points addressed: Inconsistent formatting across syllabi, High-stakes consequences for small mistakes, Front-loaded cognitive load, No single source of truth, Syllabi change mid-semester with no clear notice.

3. Solution and product design

- Solution: Allows students to upload or paste a syllabus. Uses Claude AI (Anthropic) for high-accuracy extraction and NLP methods (named entity recognition, keyword detection, date extraction) to convert unstructured course text into a clean, editable task list of assignments, exams, and deadlines.
- Design: Focuses on the highest-value user need — reducing the tedious and error-prone process of manually scanning multiple syllabi each semester. The v3 product is a fully working, single-file web app that runs directly in Chrome with no installation.
- V3 Features delivered:
 - AI-powered task extraction using Claude claude-opus-4-5 API
 - PDF upload (via PDF.js) and text paste input
 - Stats dashboard: Total Tasks, Overdue, Due This Week, Upcoming — color coded
 - Due date countdown on every task ("Due in 3 days", "Overdue 2d", "Due today!")
 - Inline edit and delete for every task card
 - Add tasks manually for anything the AI missed
 - Filter by task type (All, Exam, Assignment, Project, Quiz, Reading, Other)
 - Sort by date, type, or name
 - Export as .ics file (Google Calendar, Apple Calendar, Outlook compatible)
 - Export as CSV (includes Days Until Due column)

- Print / Save as PDF view with clean print stylesheet
- 24-hour reminder alarms embedded in .ics events
- Sample syllabus for one-click demo
- Full team credit in UI and footer
- NLP Pipeline Demo: Jupyter notebook (SyllabusSync_MVP.ipynb) demonstrates the full spaCy + dateparser pipeline covered in TAC 459 class demos, runnable in Google Colab.
- Future product vision: Google Calendar OAuth (direct push), Notion integration, deadline reminders, multi-course dashboard, syllabus change detection.

4. Metrics and success

- Measuring Success: The product's success will be evaluated through adoption, engagement, retention and accuracy metrics.
- Adoption: Track number of users, syllabi uploaded, sign-up growth. Targets: 500+ users, 1,000+ syllabi uploads, 15-20% weekly growth within first 4-6 weeks.
- Engagement: Tasks generated per user, frequency of edits, percentage exporting to calendar. Targets: 10+ tasks per user, 60%+ editing tasks, 30%+ exporting.
- Retention: Repeat usage across semesters, weekly active users. Targets: 40%+ weekly active user rate, 25%+ return within 7 days.
- Product-Specific Metrics: Extraction accuracy, edit rate, user-reported reduction in missed deadlines. Targets: 85-90% extraction accuracy, <30% edit rate, 70%+ reporting fewer missed deadlines.

5. GTM, risks and summary

- GTM: Initial target market is college students, particularly first-year students. Distribution via class group chats (Discord, GroupMe), campus Reddit communities, and student organizations. Free tool, no sign-up required, lowering barrier to entry. Launch timing: late August / early September to align with semester start.
- Risks:
 - Extraction accuracy — mitigated by using Claude AI instead of rule-based NLP, plus editable task cards so users can correct any errors
 - User trust — mitigated by transparent outputs, full editability, and countdown indicators that make errors easy to spot
 - PDF format variety — mitigated by supporting text paste as an alternative; PDF.js handles most modern PDF formats
 - API key exposure — mitigated by server-side proxy architecture in production; demo version uses grader-provided keys
- Summary: Students often struggle to extract and organize important deadlines from unstructured syllabi, leading to missed assignments and increased stress. SyllabusSync v3 is a fully functional AI-powered tool that solves this problem end-to-end — from syllabus input to Google Calendar export — built with techniques from TAC 459 and powered by the Claude API.

6. Technical design

- Architecture: Single-file web app (HTML/CSS/JavaScript) with no build step or installation required. Opens directly in Chrome. For production, a Node.js/Express backend proxies API calls so the key is never exposed client-side.
- AI Layer: Claude claude-opus-4-5 API (Anthropic) handles syllabus parsing. Prompt engineering ensures consistent JSON output. Extraction handles everything from "Due January 22" to "Feb 3" to "02/24" with automatic year inference.
- NLP Pipeline (Jupyter notebook): spaCy en_core_web_sm for Named Entity Recognition, dateparser for date normalization, regex for keyword-based task classification, pandas for structured output and CSV export. Fully runnable in Google Colab.
- Calendar Export: Standard iCalendar (.ics) format with VEVENT entries, VALARM 24-hour reminders, and DTSTART/DTEND date parsing. Compatible with Google Calendar, Apple Calendar, and Outlook.
- Privacy: No user data is stored. All processing happens in memory and is discarded after the response. No cookies, no tracking, no login required.
- PDF Parsing: PDF.js (Cloudflare CDN, v3.11.174) extracts raw text from uploaded PDFs before it enters the AI pipeline, handling a wide variety of syllabus formats.